CREDIT RISK ANALYTICS

Loan Default Prediction Project

UCI Credit Card Default Dataset

30,000 Taiwanese Credit Card Holders

Total Records	Default Rate	Best Model	AUC Score
30,000	**22.1%**	**Gradient Boosting**	**0.7803**

Linh Chi Tran

March 2026 · Python · SQL · scikit-learn · Matplotlib

1. Project Overview

Credit risk is one of the most critical challenges in banking and financial services. The ability to accurately predict whether a borrower will default on their loan enables financial institutions to make better lending decisions, price risk appropriately, and manage portfolio exposure.

This project builds a complete end-to-end Loan Default Prediction system using machine learning applied to the UCI Credit Card Default Dataset — a well-established academic benchmark used in credit risk research worldwide.

1.1 Objectives

- Build a production-grade SQL database schema to store borrower and credit card data
- Engineer meaningful features from raw 6-month payment history and billing data
- Train and compare three machine learning models to predict default probability
- Assign risk grades (A–F) to all 30,000 borrowers based on model output
- Generate an interactive visual dashboard to communicate findings

1.2 Dataset Description

Attribute	Details
Source	UCI Machine Learning Repository / Kaggle
Records	30,000 credit card clients in Taiwan
Features	25 columns: demographics, credit limit, 6-month payment status, bill & payment amounts
Target Variable	default.payment.next.month (1 = will default, 0 = will not)
Period	April 2005 – September 2005
Class Balance	77.9% No Default / 22.1% Default

2. Exploratory Data Analysis

2.1 Default Rate by Education Level

Education level shows a strong relationship with default probability. High School graduates carry the highest risk at 25.2%, while Graduate-level borrowers default at a lower rate of 19.2%. This insight can inform credit limit decisions during loan origination.

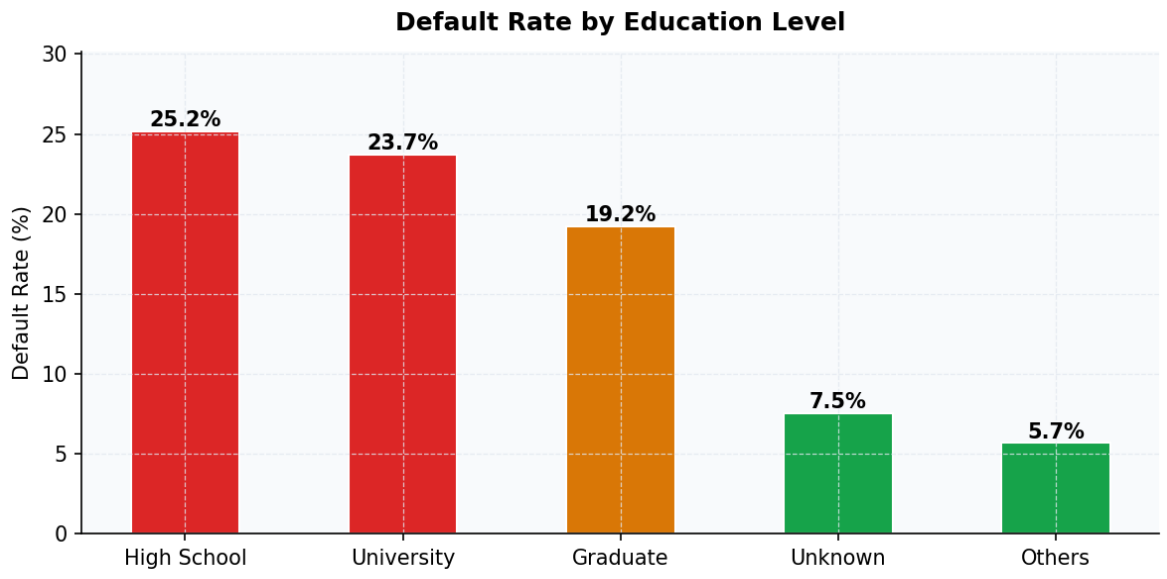

Figure 1 — Default Rate by Education Level. Red = High Risk (>23%), Amber = Medium, Green = Lower Risk.

2.2 Default Rate by Age Group

Borrowers in their 20s show the highest default rate, likely due to lower income stability and less credit experience. Default rates decline steadily through the 30s and 40s before rising again in the 60+ group, possibly due to fixed income constraints.

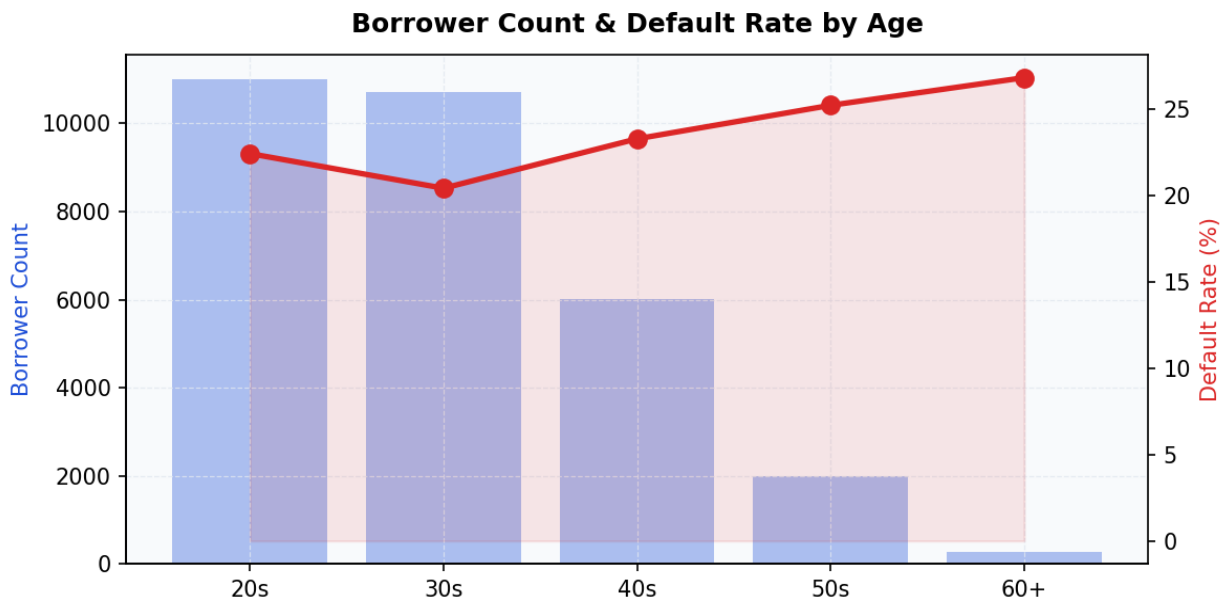

Figure 2 — Loan volume (bars) and default rate (line) by age group.

2.3 Default Rate by Credit Utilization

Credit utilization rate (average bill amount divided by credit limit) is one of the strongest engineered features. Borrowers using 80–100% of their credit limit default at nearly twice the rate of those using less than 20%, confirming that high utilization is a key stress indicator.

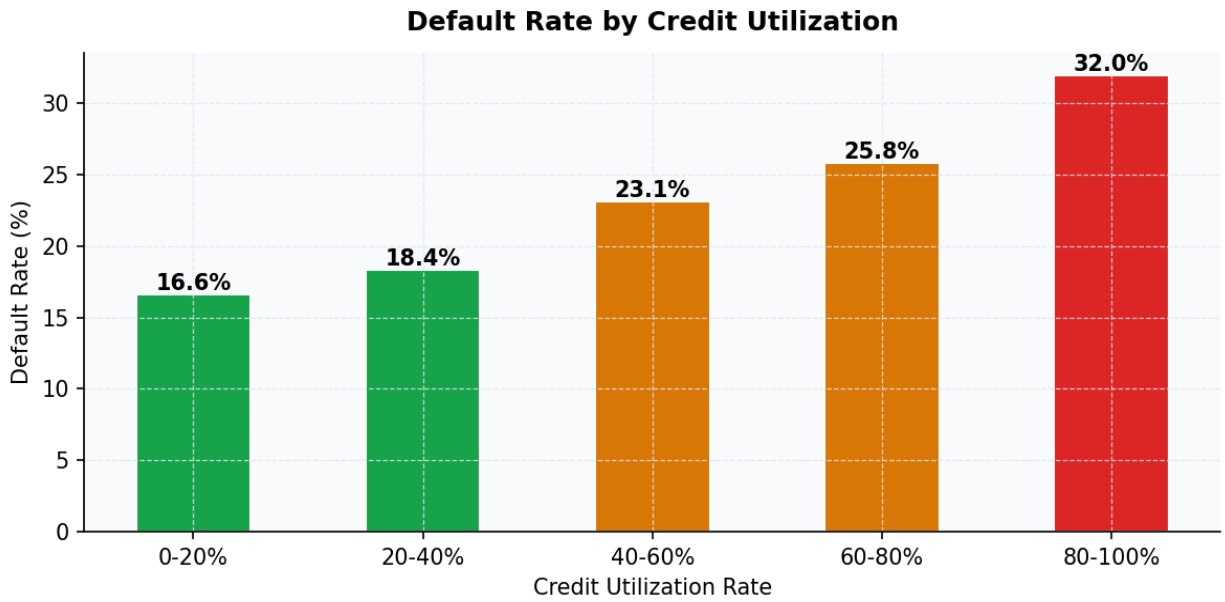

Figure 3 — Default rate rises sharply with credit utilization. Borrowers above 60% utilization warrant closer scrutiny.

2.4 Credit Limit Distribution

Defaulters tend to have lower credit limits on average, suggesting that higher-limit clients are either more creditworthy at origination or have stronger financial capacity to service their debt.

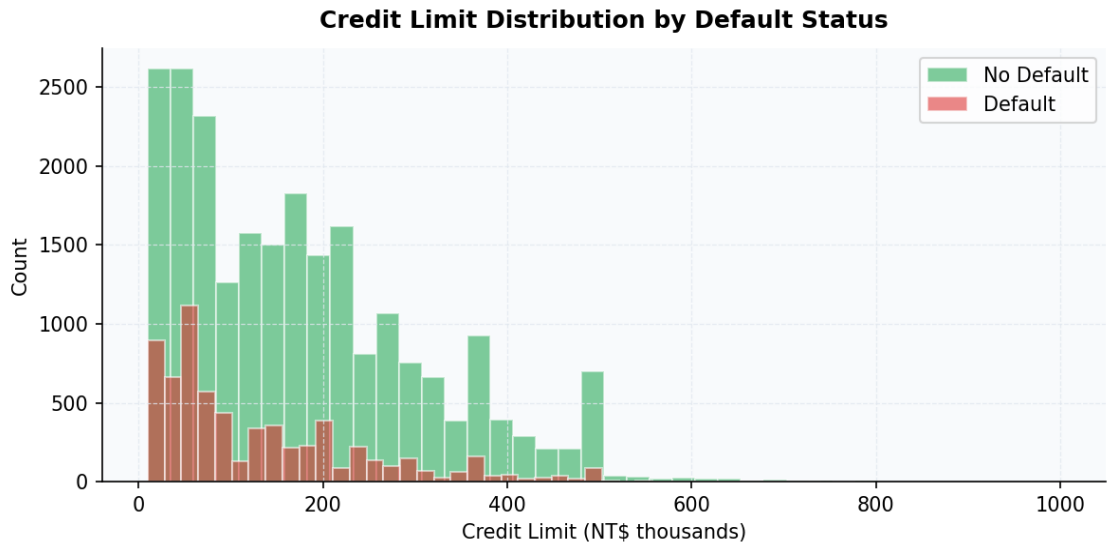

Figure 4 — Credit limit distribution for defaulters vs non-defaulters. Defaulters cluster at lower limits.

3. Methodology & Pipeline

3.1 Technical Architecture

Step	Script	Purpose	Output
1	**load_data.py**	Parse CSV, engineer features, load SQLite	credit_risk.db
2	**train_model.py**	Train 3 ML models, evaluate, select best	model_artifacts.pkl
3	**score_loans.py**	Score 30,000 borrowers with risk grades A–F	loan_risk_scores table
4	**dashboard.py**	Generate 10-chart HTML visual dashboard	dashboard.html

3.2 Engineered Features

Feature	Formula	Significance
utilization_rate	avg_bill_amt / credit_limit	Key stress indicator — higher = more risk
months_late	Count of pay_status columns > 0	Strongest single default predictor
total_payments	Sum of 6 monthly payment amounts	Measures payment capacity
avg_bill_amt	Mean of 6 monthly bill amounts	Average monthly credit exposure

4. Model Results

4.1 Model Comparison

Model	ROC-AUC	Accuracy	Avg Precision	Brier Score
Logistic Regression	0.7439	76%	0.5016	0.1932
Random Forest	0.7769	78%	0.5570	0.1737
Gradient Boosting	**0.7803**	**82%**	**0.5630**	**0.1344**

4.2 ROC Curve — Gradient Boosting

The ROC curve measures the model's ability to distinguish between defaulters and non-defaulters across all classification thresholds. An AUC of 0.7803 means the model correctly ranks a randomly chosen defaulter above a non-defaulter 78% of the time — well above the 0.5 baseline of random guessing.

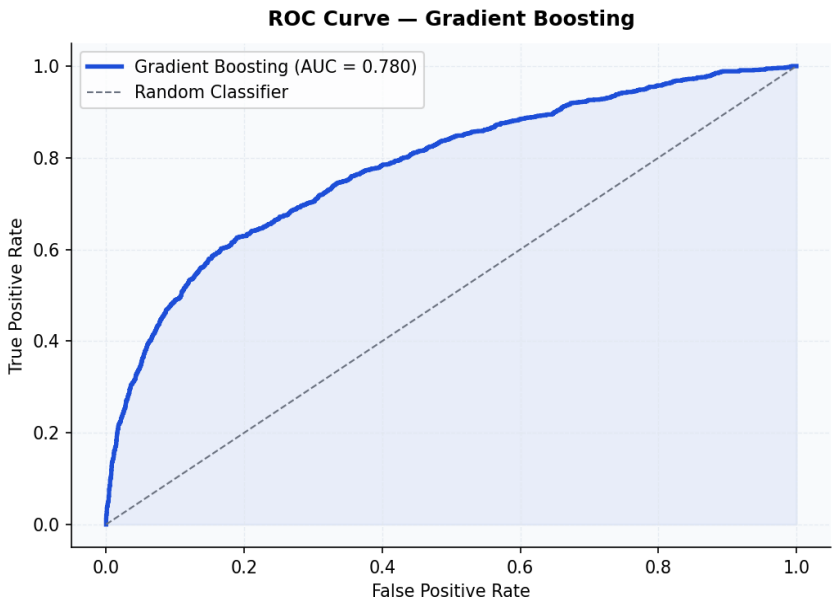

Figure 5 — ROC Curve for Gradient Boosting (AUC = 0.7803). The blue shaded area represents model performance above random

4.3 Precision-Recall Curve

The Precision-Recall curve is especially relevant for imbalanced datasets. An Average Precision of 0.5630 indicates the model maintains reasonable precision even as recall is pushed higher — important when the cost of missing a true defaulter is high.

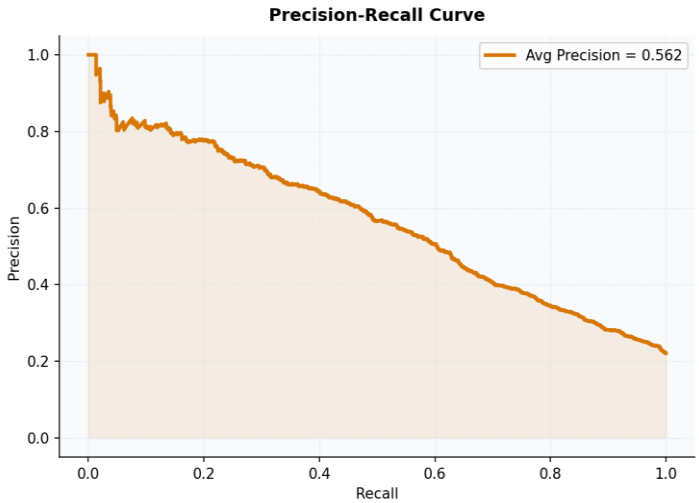

Figure 6 — Precision-Recall Curve (Average Precision = 0.5630). Higher area under the curve = better default detection.

4.4 Model Score Distribution

The score distribution shows clear separation between defaulters and non-defaulters, validating the model's discriminative power. The overlap zone around 0.3–0.6 represents the most uncertain borrowers where manual review may be warranted.

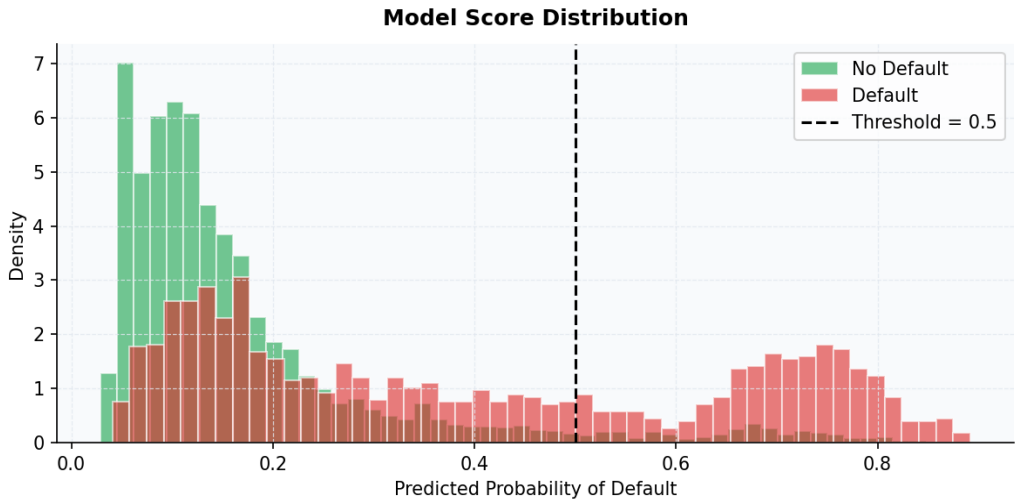

Figure 7 — Predicted default probability for test set borrowers. Green = No Default, Red = Default.

4.5 Top 15 Feature Importances

The feature importance chart reveals which variables the Gradient Boosting model relies on most. Payment status variables (months_late, pay_status_sep) dominate, confirming that recent payment behavior is the strongest signal for predicting default.

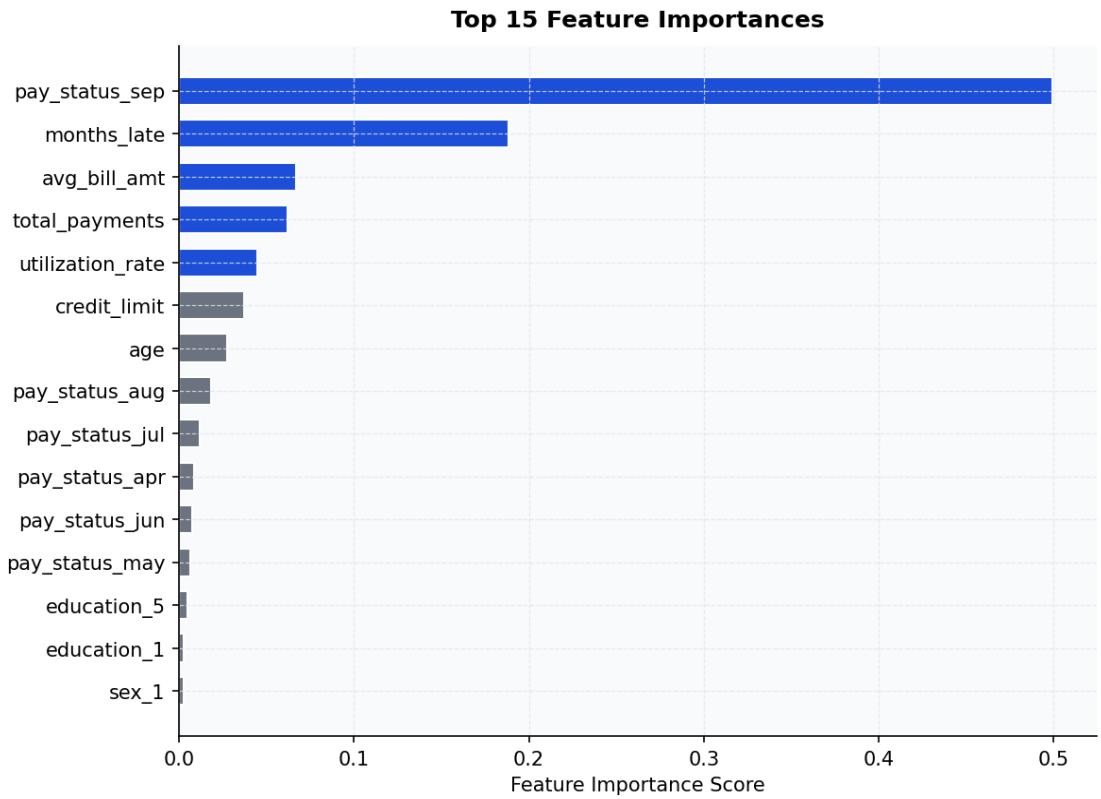

Figure 8 — Top 15 features by importance score. Blue bars indicate the most influential features.

5. Risk Scoring Results

5.1 Risk Grade Distribution

After training, the model scored all 30,000 borrowers and assigned each a risk grade from A (safest) to F (highest risk) based on their predicted probability of default. The majority of borrowers fall into Grade B or C, reflecting a moderately risky portfolio overall.

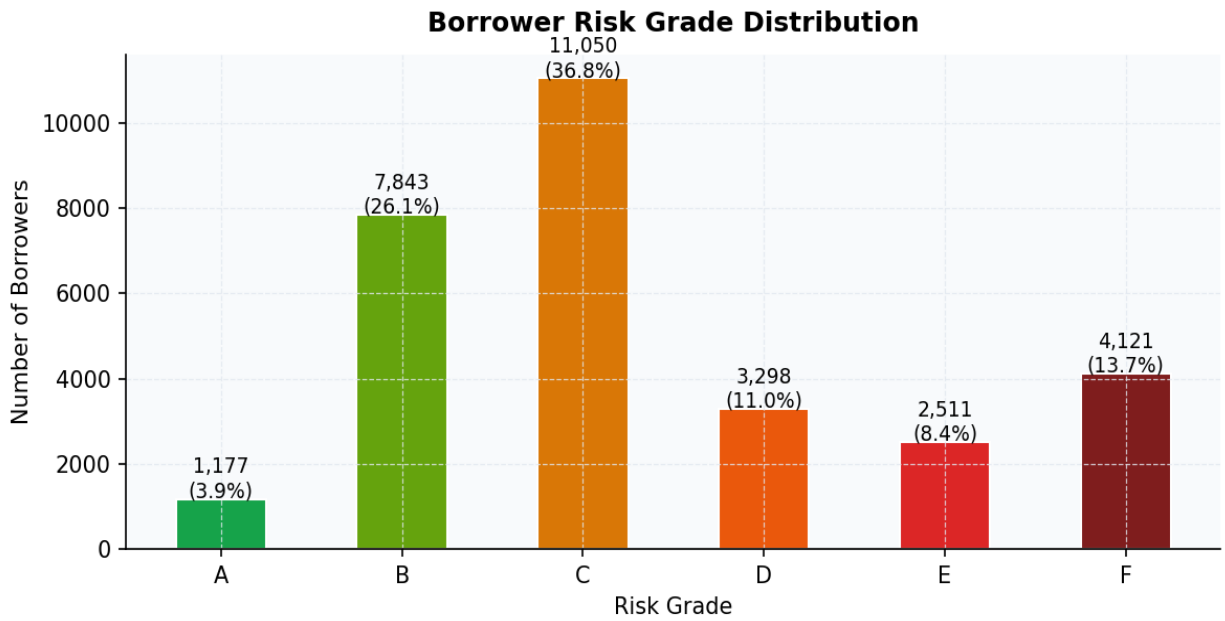

Figure 9 — Distribution of 30,000 borrowers across risk grades A–F.

Grade	Risk Level	Count	Share	PD Threshold
A	Very Low Risk	1,168	3.9%	PD < 5%
B	Low Risk	7,845	26.1%	5% ≤ PD < 10%
C	Moderate Risk	11,049	36.8%	10% ≤ PD < 20%
D	Elevated Risk	3,317	11.1%	20% ≤ PD < 30%
E	High Risk	2,493	8.3%	30% ≤ PD < 45%
F	**Very High Risk**	4,128	13.8%	PD ≥ 45%

6. Conclusions & Recommendations

6.1 Key Findings

- Gradient Boosting achieved the best ROC-AUC of 0.7803, outperforming Logistic Regression and Random Forest
- months_late and pay_status_sep are the strongest default predictors — recent payment behavior matters most
- High School graduates default at 25.2%, significantly higher than graduates (19.2%)
- Borrowers in their 20s carry the highest default risk across all age groups
- High credit utilization (>80%) doubles the default rate compared to low utilization (<20%)
- 22.1% of borrowers defaulted — indicating a high-risk portfolio overall
- 13.8% of borrowers (4,128) were assigned Grade F with predicted default probability above 45%

6.2 Recommendations

- Lower the classification threshold from 0.5 to 0.3 to catch more true defaulters (improves recall)
- Apply stricter credit limits to borrowers aged 20–29 or with High School education only
- Flag borrowers with utilization > 80% for proactive outreach before default occurs
- Consider SMOTE oversampling or XGBoost to further improve minority class detection
- Retrain model periodically as macroeconomic conditions change over time

6.3 Technology Stack

Component	Technology	Purpose
Database	SQLite + SQL	Schema, data storage, analytical views
Data Processing	Python + Pandas	CSV parsing, feature engineering, data cleaning
Machine Learning	scikit-learn	Logistic Regression, Random Forest, Gradient Boosting
Visualization	Matplotlib + HTML	9 analytical charts embedded in this report
Environment	Python 3.9 + Anaconda	macOS, VS Code IDE

End of Report — Credit Risk Analytics Project | March 2026